

Q1. Suppose `vector<int>` is a max heap. Write following member function:

```
void modify(int i, int newKey){ //i is a valid index and newKey can be larger or smaller than existing key
    int oldKey = val[i]; //val is a vector, where keys are stored
    heap[i] = newKey;
    if (newKey > oldKey)    heapifyUp(i); //the function compare with parent and swap
    else if (newKey < oldKey) heapifyDown(i); //the function compare with child nodes and swap
```

Q2. Write sort functions without comparing the elements, it is guaranteed that elements are in range 500 to 800 only.

```
void sort(vector<int>& v){
    const int RANGE = 800 - 500 + 1;
    vector<int> frequency(RANGE, 0);
    for (int num : v)
        frequency [num - 500]++;
    for (int i = 0, index = 0 ; i < RANGE; i++)
        for ( ; frequency[i] > 0 ; frequency [i]--)
            v[index++] = i + 500;
}
```